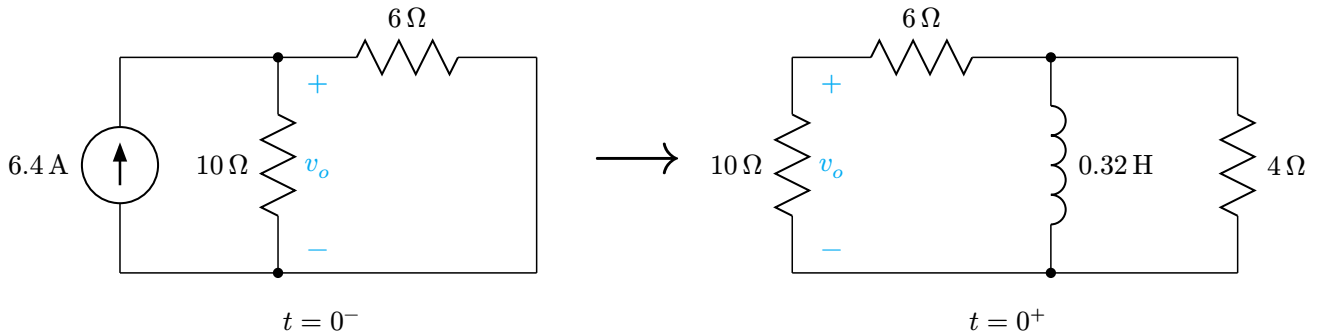


Lab 1: First-Order (RL/RC) Circuits

No. 1

Step 1. Find $v_o(t)$ for $t \geq 0^+$



for $t = 0^-$:

$$10 \parallel 6 = 3.75\ \Omega$$

$$v_o(0^-) = 6.4(3.75) = 24\text{ V}$$

$$i_L(0^-) = \frac{24}{6} = 4\text{ A}$$

for $t > 0$:

$$i_L(0^+) = i_L(0^-) = 4\text{ A}$$

$$R_{\text{eq}} = 4 \parallel (10 + 6) = 3.2\ \Omega$$

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{0.32}{3.2} = 0.1\text{ s}$$

Note that $i_L(\infty) = 0$ for natural response

$$i_L(t) = \cancel{i_L(\infty)}^0 + [i_L(0^+) - \cancel{i_L(\infty)}^0] e^{-t/\tau}$$

$$i_L(t) = i_L(0^+) e^{-t/\tau}$$

$$\boxed{i_L(t) = 4e^{-10t}\text{ A}} \text{ for } t \geq 0$$

$$i_o(t) = \frac{4}{16 + 4} i_L(t) = 0.8e^{-10t}$$

$$-v_o(t) = 10i_o(t)$$

$$\boxed{v_o(t) = -8e^{-10t}\text{ V}} \text{ for } t \geq 0$$

Step 2. Simulate the circuit and confirm the result in Step 1

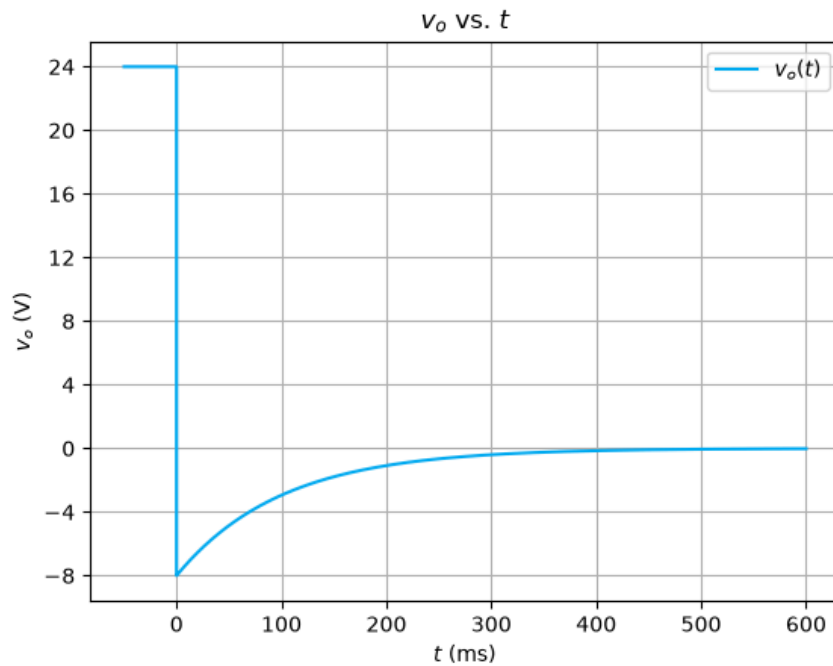


Figure 1: The simulated graph for $t > 0$ closely matches the theoretical function $v_o(t) = -8e^{-10t}$ V

Step 3. Find the time at which 75% of the initial stored energy has been dissipated.

$$w(t) = \frac{1}{2}Li^2(t)$$

$$i(t) = 2.56e^{-20t} \text{ A}$$

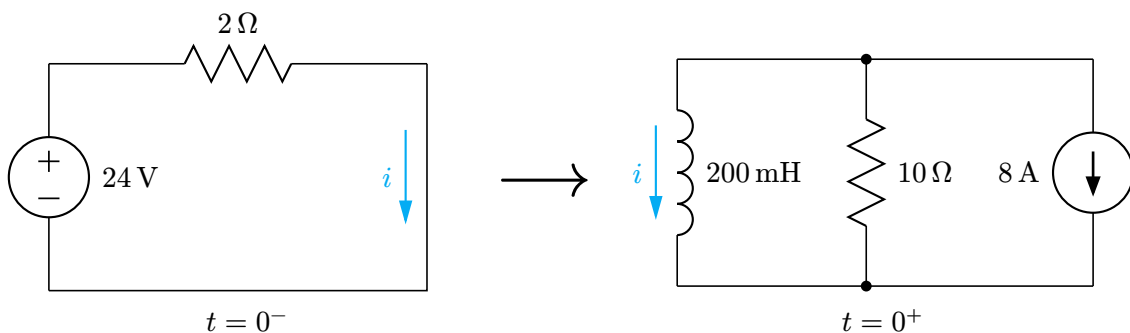
$$(1 - 0.75)w(0) = 2.56e^{-20t}$$

$$\Rightarrow t = -\frac{1}{20} \ln\left(\frac{0.25w(0)}{2.56}\right)$$

$$t = 0.069 \text{ s}$$

No. 2

Step 1. Find $i(t)$ for $t \geq 0^+$



for $t = 0^-$:

$$i(0^-) = \frac{24}{2} = 12 \text{ A}$$

for $t > 0$:

$$i(0^+) = i(0^-) = 12 \text{ A}$$

$$R_{\text{eq}} = 10 \Omega$$

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{0.2}{10} = 0.02 \text{ s}$$

As $t \rightarrow \infty$, the inductor shorts and all 8 A from the current source flows through it. Since i is defined in the opposite direction as the source, we flip the sign.

$$i(\infty) = -8 \text{ A}$$

$$i(t) = i(\infty) + [i(0^+) - i(\infty)]e^{-t/\tau}$$

$$i(t) = -8 + [12 - (-8)]e^{-t/0.02}$$

$$\boxed{i(t) = -8 + 20e^{-50t} \text{ A}} \text{ for } t \geq 0$$

Step 2. Simulate the circuit and confirm the result in Step 1

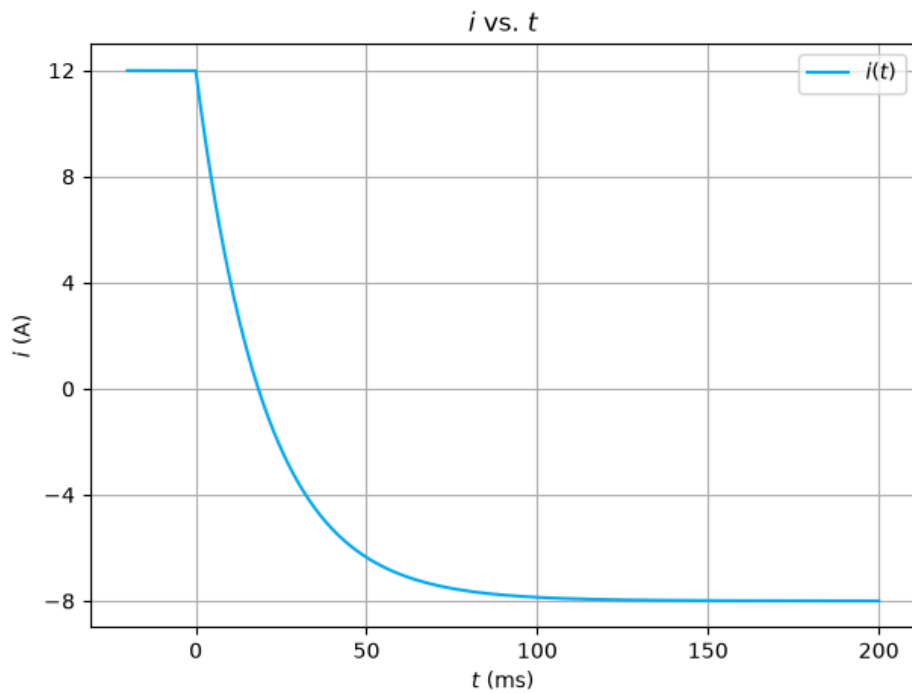
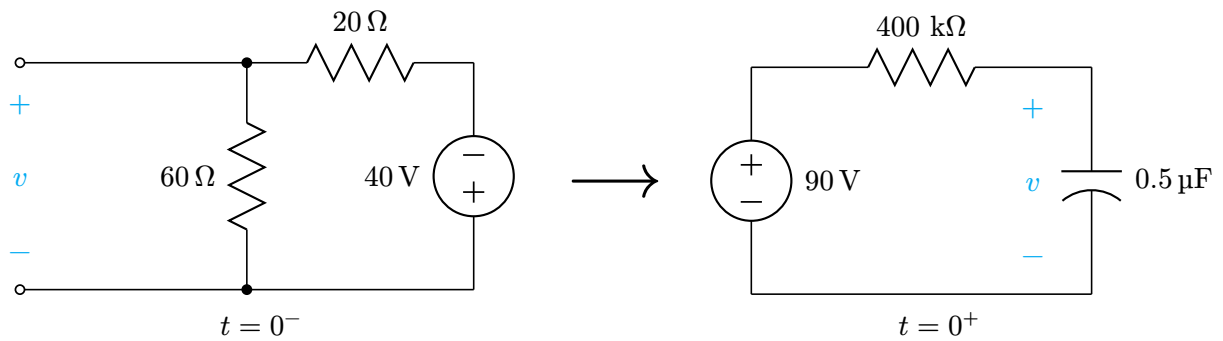


Figure 2: The simulated graph for $t > 0$ closely matches the theoretical function

$$v_o(t) = -8 + 20e^{-50t} \text{ A}$$

No. 3

Step 1. Find $v(t)$ for $t \geq 0^+$



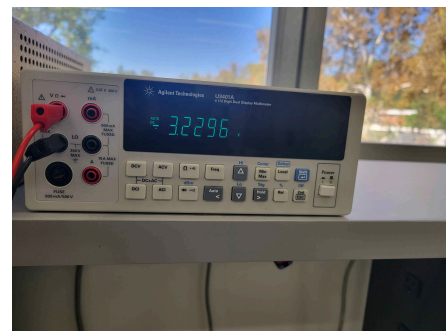
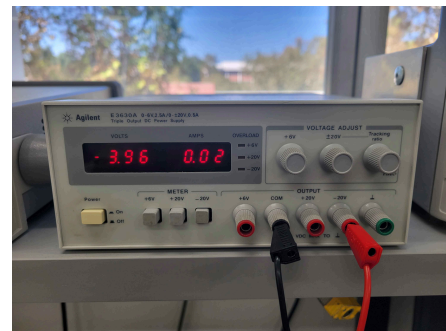
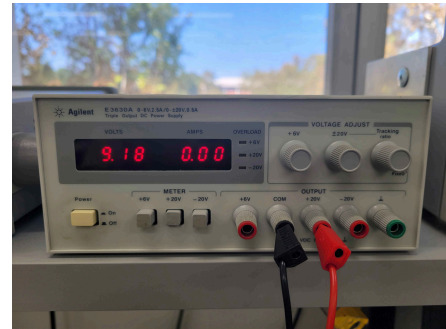
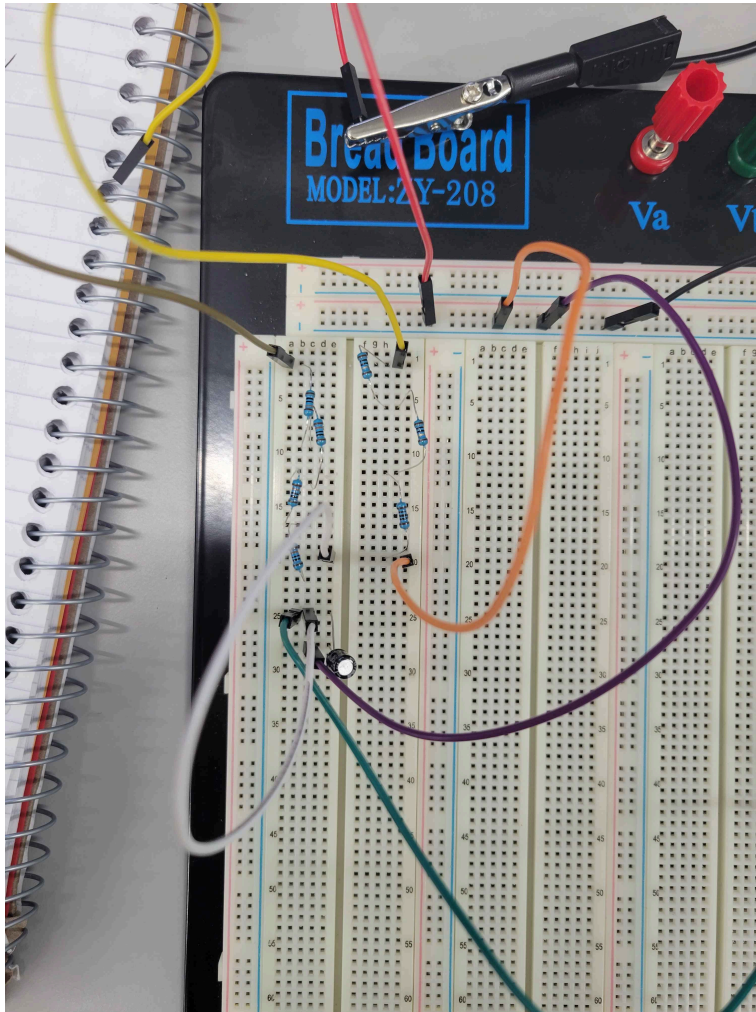
for $t = 0^-$:

$$i = \frac{40}{60 + 20} = 0.5\text{ A}$$
$$-v(0^-) = 60i$$
$$\Rightarrow v(0^-) = -30\text{ V}$$

for $t > 0^+$:

$$v(0^+) = v(0^-) = -30\text{ V}$$
$$R_{\text{eq}} = 400\text{ k}\Omega$$
$$\tau = R_{\text{eq}}C = (400\text{ k}\Omega)(0.5\ \mu\text{F}) = 0.2\text{ s}$$
$$v(\infty) = 90\text{ V}$$
$$v(t) = v(\infty) + [v(0^-) - v(\infty)]e^{-t/\tau}$$
$$v(t) = 90 + [-30 - 90]e^{-t/0.2}$$
$$\boxed{v(t) = 90 - 120e^{-5t}\text{ V}} \text{ for } t \geq 0$$

Step 2. Implement the circuit on a breadboard and confirm the result in Step 1



Step 3. Simulate the circuit and confirm the result in Steps 1-2.

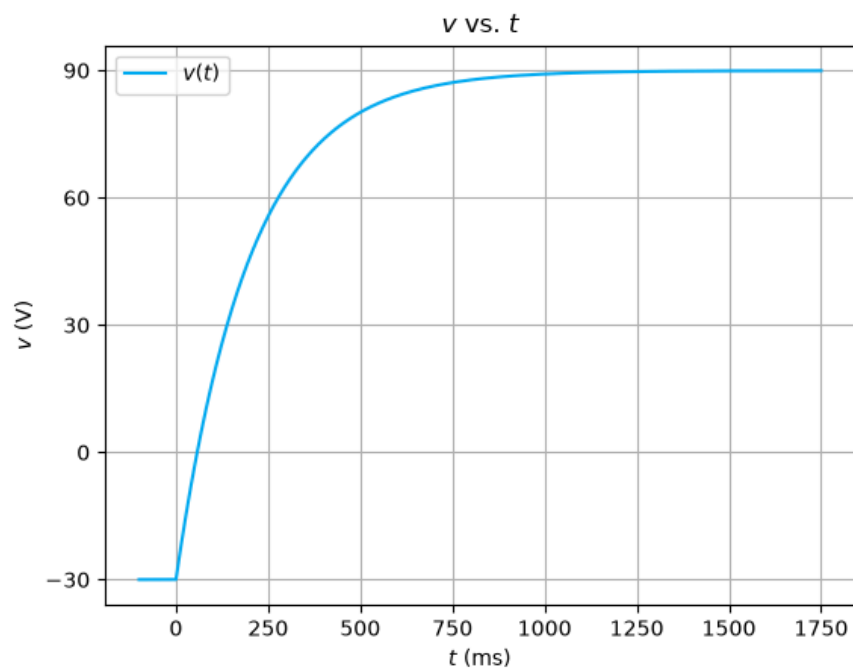


Figure 3: The simulated graph for $t > 0$ closely matches the theoretical function $v_o(t) = 90 - 120e^{-5t}$ V